

Quarry Sub-Sector Profile



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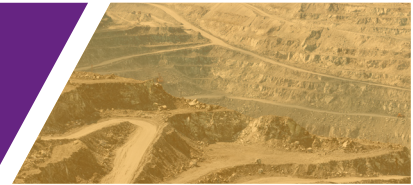
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Message from the Quarry Sub-Sector Chair



Kenya's Quarry Sub-Sector plays a strategic role in our economy, since majority of buildings are constructed using raw materials derived from the earth, through quarrying.

As of 2019, there were approximately 160 aggregates quarries in the country, out of which 68 were in Machakos and Kitui Counties. Combined, they directly employ over 28,797.

Stone mining and quarrying primarily include the development of mine sites, mining and quarrying aggregates obtained from crushed stone and dimension stone such as building blocks. It also comprises beneficiation activities such as crushing, grinding, washing, screening, pulverizing (reduce to dust or powder) and sizing.

Kenya's quarry products, include aggregates from basalt, phonolite and trap rock; dimension stone, including building stone, marble and granite decorative slabs; and limestone and pozzollana for cement production. When pulverized, limestone produces agricultural lime.

The construction industry depends on quarried products in order to thrive. Additionally, infrastructural development links directly to economic growth. Roads, railways, ports, dams, main power transmission lines and buildings, commercial, institutional and residential, require large volumes of aggregates - they rely heavily on well-grounded aggregates production.

We laud the government's initiatives, geared towards driving the Quarry Sub-Sector's growth. However, some challenges abound. KAM remains committed to engaging our partners and stakeholders, to resolve these challenges, by implementing the recommendations made in this profile.

Dhiraj Patel
Ag. Chair KAM Quarry Sub-sector



Message from the CEO



The Quarry Sub-Sector Profile gives an in-depth analysis of KAM's Quarry Sub-Sector, which is rapidly expanding due to ongoing major infrastructural projects across the country.

Globally, the quarry industry is growing at a fast pace, because of the ever-increasing demands for its products. This demonstrates the quarry industry's potential, to contribute more to the country's GDP, as well as economic development. Therefore, for our economy to grow, we need more aggregates, which are the foundation of our nation's infrastructure.

Here in Kenya, the Quarry Sub-Sector has links to a wide variety of economic pillars, including real estate and transport and infrastructure, among others. The current government's prioritization of Manufacturing and Affordable Housing under the Big Four Agenda, gave it the much-needed shot in the arm, to spur its growth.

However, more needs to be done to further enable the Quarry Sub-Sector reach its full potential. The sector continues to face challenges in royalty fees, high cost of power, import duty and VAT on machinery and machinery parts, double taxation by the National and County Governments, unclear demarcation and zoning of commercial and residential areas leading to conflicts and inadequate skills.

Kenya Association of Manufacturers recommends the formation of a taskforce comprising of the Quarry Sub-sector, Ministry of Petroleum and Mining and its relevant agencies and Council of Governors, to resolve these issues.

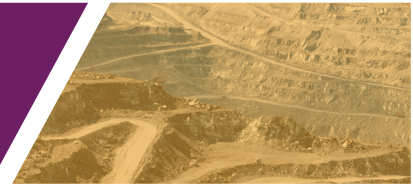
The Quarry Sub-Sector is the building block of our society. Its stability and success are paramount for our nation's shared prosperity.



...for our economy to grow, we need more aggregates, which are the foundation of our nation's infrastructure.

Phyllis Wakiaga
KAM Chief Executive

1 Introduction



The bulk of our built environment is formed from the raw materials derived from the earth by quarrying. If the raw materials were not available, life would be more basic and less comfortable for infrastructural development, including building and construction. Stone quarrying is the multi stage process by which rock is extracted from the ground and crushed to produce aggregates, which is then screened into required sizes for immediate use, or further processing such as pulverization for lime production or coating with bitumen to make asphalt for road construction. The most common ore used to make crushed stone includes basalt, phonolite, granite, limestone, trap rock, dolomite and sandstone.

Stone mining and quarrying primarily include the development of mine sites, mining and quarrying aggregates obtained from crushed stone and dimension stone such as building blocks. It also comprises beneficiation activities such as crushing, grinding, washing, screening, pulverizing (reduce to dust or powder) and sizing.

2 Global Quarrying Industry

Globally, the quarry industry is gaining momentum due to the immense demand for its products and large volume of infrastructural development. Examples include real estate in major cities and towns, road network expansions, train project constructions, ports expansions and other demanding development, thus pushing the sector to grow rapidly, in order to sustain market demand. At global level, the industry is still fragmented. The top 20 global aggregates producers comprise only 5% of production, with the remaining 95% being produced by small and medium scale enterprises.

Empirical data indicates that there is a direct relationship between tons per capita and the GDP. This means that more aggregates are needed for the development of the economy. For instance, in the USA, the quarry industry has contributed about 3.1% over the last five years to the economy and it is steadily growing. In China, the overall consumption is 20 billion per year and it is expected to rise by a double multiplier.

The global stone mining and quarrying sector can be segmented based on stone, type, application and region. In terms of stone, the industry can be divided into marble, granite, limestone,

sandstone and others. The marble segment is anticipated to expand rapidly as its application in building is growing at a significant pace. On the other hand, the sandstone segment will also witness a significant growth owing to the rise in demand for aggregates used in the construction industry.

In terms of type of quarries the industry can be divided into dimension stone mining and crushed stone mining. The crushed stone mining segment is expected to witness significant growth, owing to the rise in demand for aggregates used in construction of building foundations, railroad beds and highways. Crushed stone quarrying is further divided into different categories of operations namely:

Single bench quarry with spiral access

Multiple bench quarry on a hillside

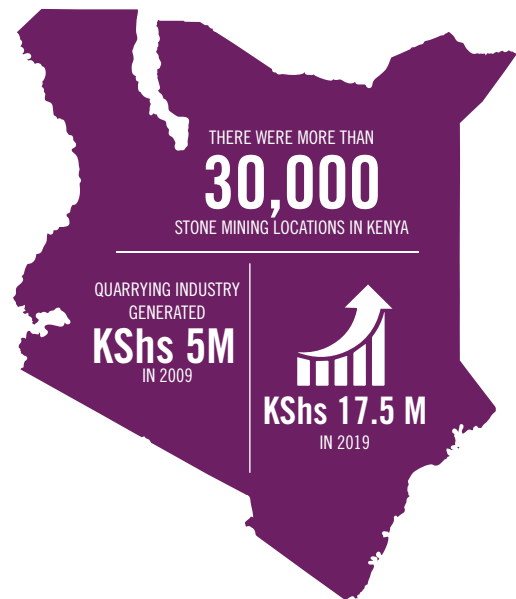
Multiple bench subsurface quarry

Subsurface quarry with switch back access



3 Quarrying in Kenya

Like the rest of developing countries, the quarrying industry in Kenya has been rising due to demand for building materials occasioned by population growth, urbanization and expanding economy. According to the Kenya National Bureau of Statistics (KNBS, 2009), there were more than 30,000 stone mining locations in Kenya. The figure is on the rise, the quarrying industry generated KShs 5 million in June 2009, and rose to KShs 17.5 million in 2019, as indicated by data from CIEC.. The results indicate that the revenue has tripled, which is an indication that the number of mines and quarries in the country has increased.



The products of quarrying in Kenya include aggregates from basalt, phonolite and trap rock, dimension stone including building stone, marble and granite decorative slabs; and limestone and pozollana for cement production. Limestone when pulverized produces agricultural lime.

Records from the Department of Mining show that there are about 160 aggregates quarries in the country as at 2019, out of which 68 quarries are in Machakos and Kitui Counties and combined, directly employ over 28,797 people.

4 Economic Impacts of Stone Quarrying

The construction industry depends on quarried products in order to thrive. The infrastructural development links directly to economic growth. Roads, railways, ports, dams, main power transmission lines and buildings, both commercial, institutional and residential require large volumes of aggregates and therefore rely heavily on well grounded aggregates production.



4.1 Contribution of Quarrying to the Kenyan Economy

The table below shows the construction demand trends in the country, as provided by the 2019 KNBS Economic survey report. In 2018 the construction industry registered a slow growth of 6.3% compared to 8.5% in 2017.

Table 1: Selected indicators in the constructions sector during 2014-2018

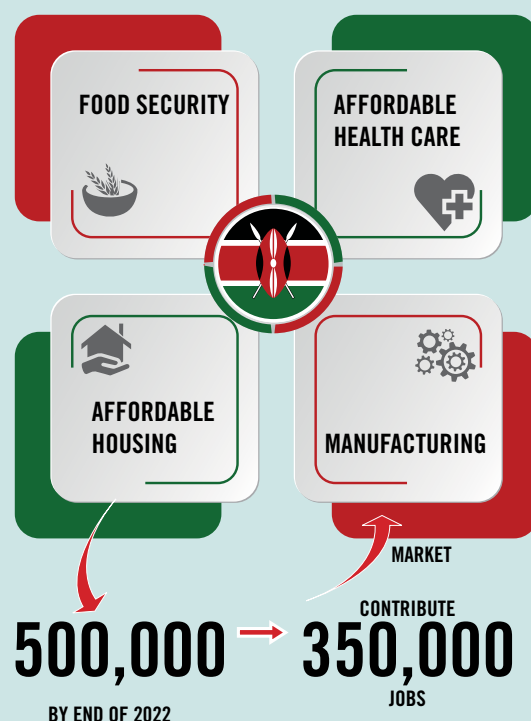
Indicator	2015	2016	2017	2018	2019*
Index of reported private building works completed in Nairobi City County.	369.4	409.3	443.1	466.2	482.3
Index of reported public building works completed in major towns.	112.6	69.21	59.4	57.9	39.0
Index of government expenditure on roads.	350.3	462.8	388	415.4	456.8
1 Index of Employment.	245.0	269.9	277.1	284.1	286.9
Cement consumption ('000 tonnes)	5,708.8	6,310.1	5,857.9	5,948.7	5,933.3
Private Employment ('000 persons)	140.2	155	159.4	163.0	164.5
Public Employment ('000 persons)	7.9	8	8.5	8.6	8.8

Source: KNBS, 2020

4.2 The Big 4 Agenda and the Mining Sector

The Government has initiated the implementation of the Big 4 Agenda to be achieved by the end of 2022. In this plan the Government aims at ensuring “food security, affordable health care, affordable housing and manufacturing” by the end 2022.

The affordable housing and manufacturing are connected with the quarry sub sector. Because the government intends to build low cost houses, the quarry sub sector is the source of building stones, aggregates as well as other construction materials. The government intends to build more than 500,000 affordable cost houses by the end of 2022. This will require tons of building stones, aggregates and other construction materials that directly come from the quarry sub sector. To reduce the construction cost, the government will need to negotiate for low rates for key construction inputs for the participating developers. It is estimated that the real estate and construction sector will contribute 350,000 jobs and provide a market for manufacturers and suppliers apart from their contribution to the country's GDP.



5 Geology of Quarrying Rocks and their uses in Kenya

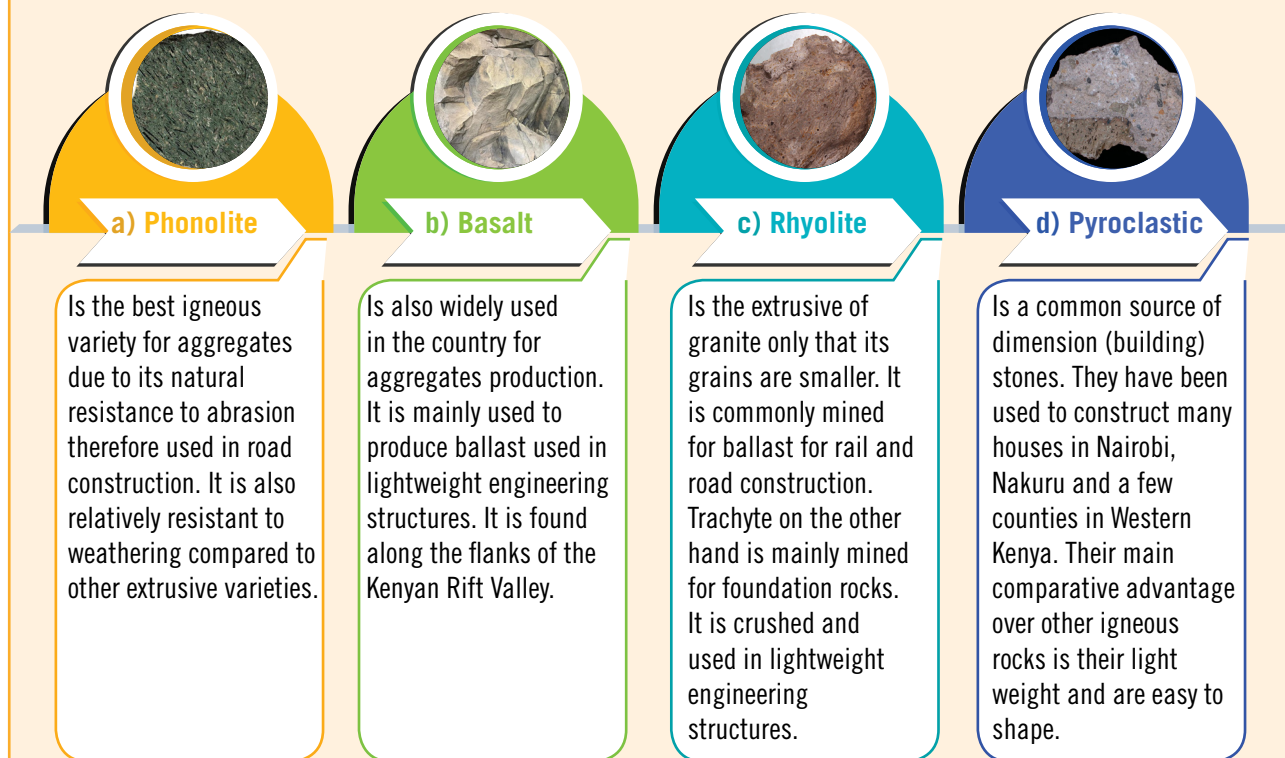
The geology gives the information necessary for preliminary consideration for locating a quarry. The rock suitability and the quantity is key in this determination.

Kenya is endowed with three categories of rocks namely:



A. Igneous rocks are further divided into two categories that **is extrusive** and **intrusive igneous** rocks.

i. Extrusive igneous rocks are quite extensive in many parts of the country except North-Eastern. Several of these rock types have been processed for different purposes. The commonly used volcanic rocks include:



ii. Intrusive Igneous

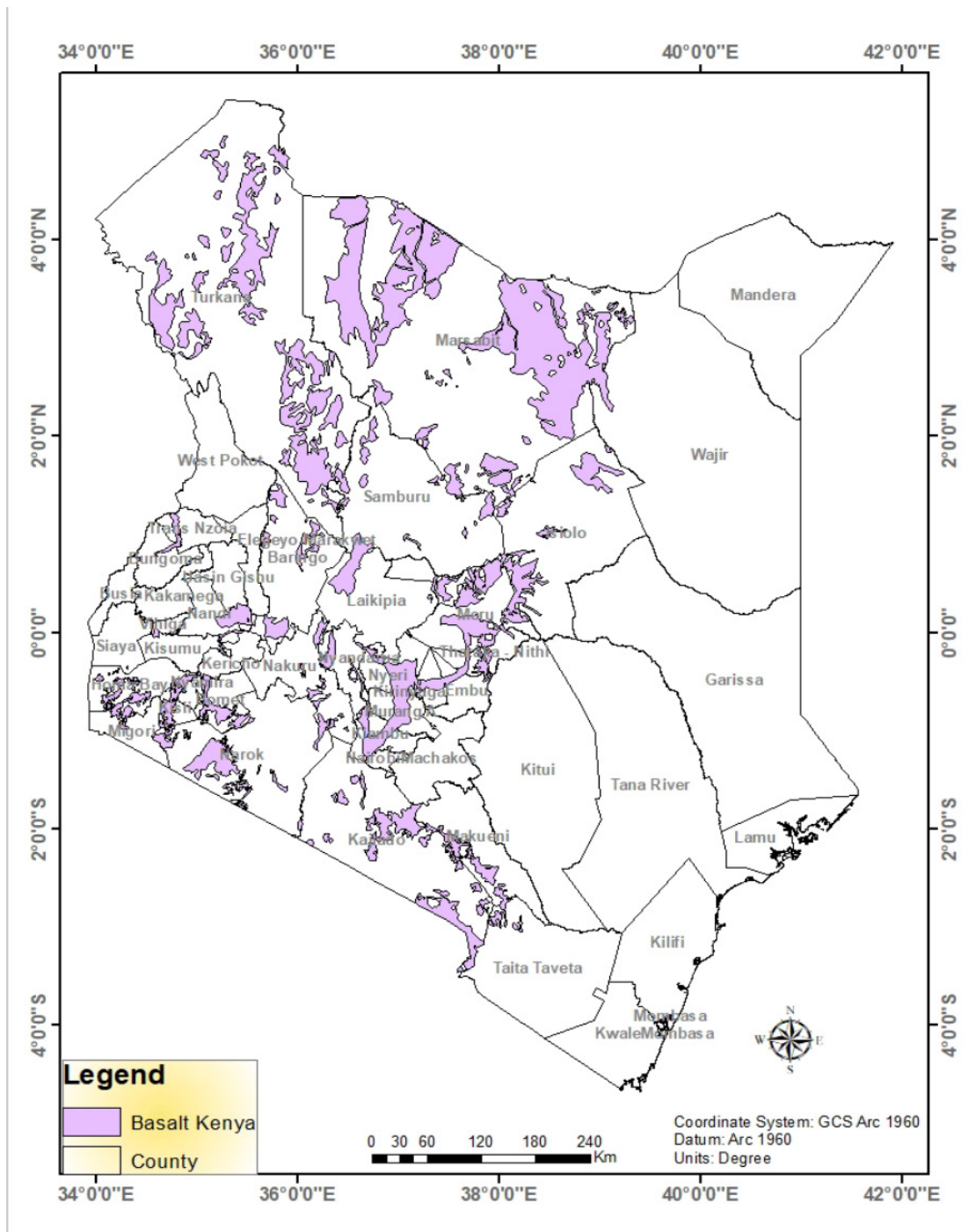
Intrusive igneous rock on the other hand is formed deep within the earth's crust. Their major characteristic is the huge crystals compared to extrusive igneous rock. In most cases they are relatively harder than the extrusive types. They are, therefore, more preferred for specific purposes.

Granite is the most common in this category, and is mostly used as dimension stone for its aesthetic qualities. However, it is also used as aggregates in road construction.



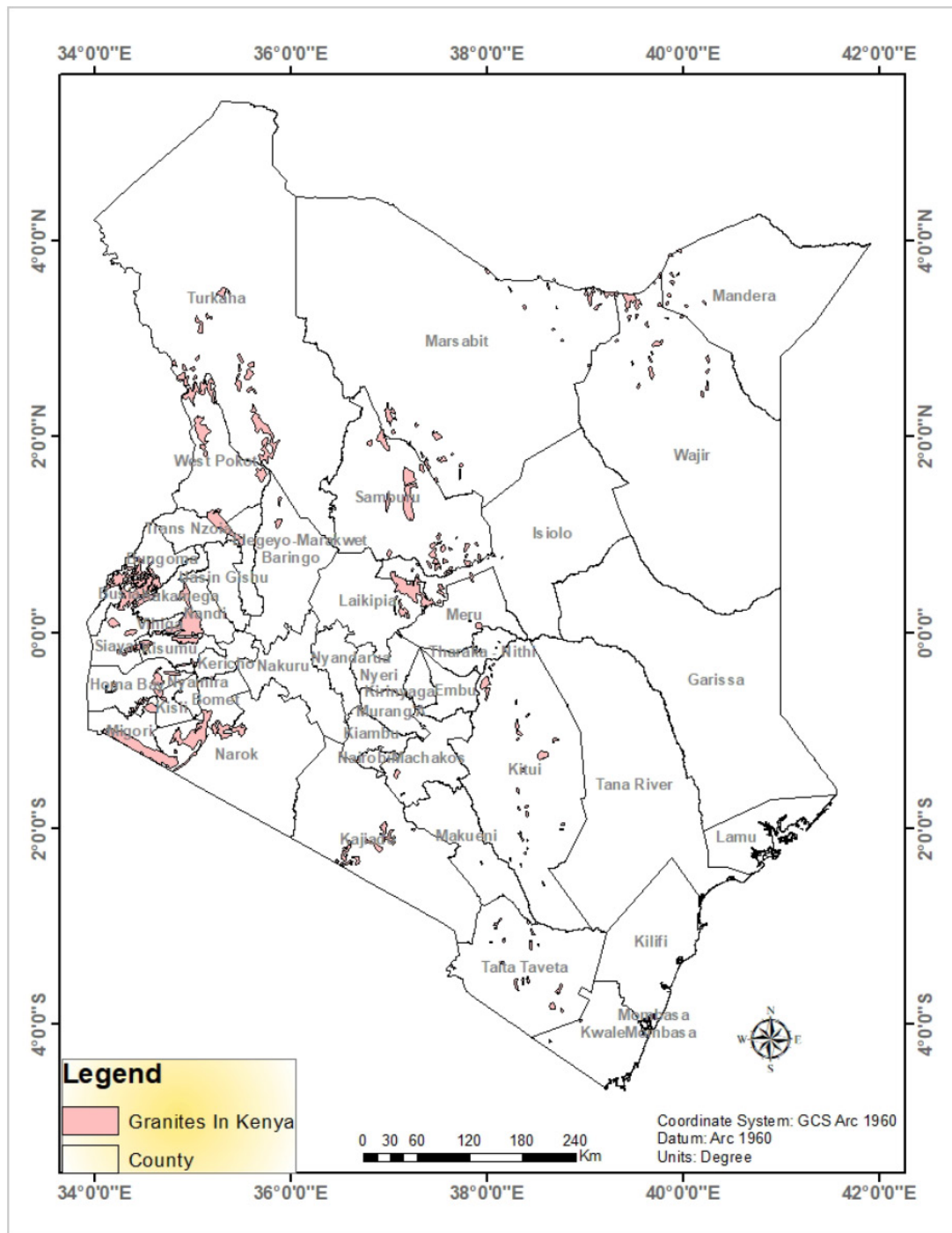
B. Metamorphic Rock: Metamorphic rock is made from thermal heating of either igneous or sedimentary rocks until complete or partial melting. They are generally found within the Mozambique belt. They are formed below the other two rocks. This type of rock is mainly exposed in the following counties of Kenya - Kitui, Machakos, Makueni, Taita Taveta, Tharaka Nithi, Elgeyo Marakwet and some parts of West Pokot. Most of these rocks are used for their aesthetic qualities, the major one being granite gneiss.

C. Sedimentary Rocks: Sedimentary rocks are formed from eroded rocks, while others are formed from dissolved and evaporated minerals. The key sediments in quarrying industry include sand and dimension stones found in areas around coastal Kenya.



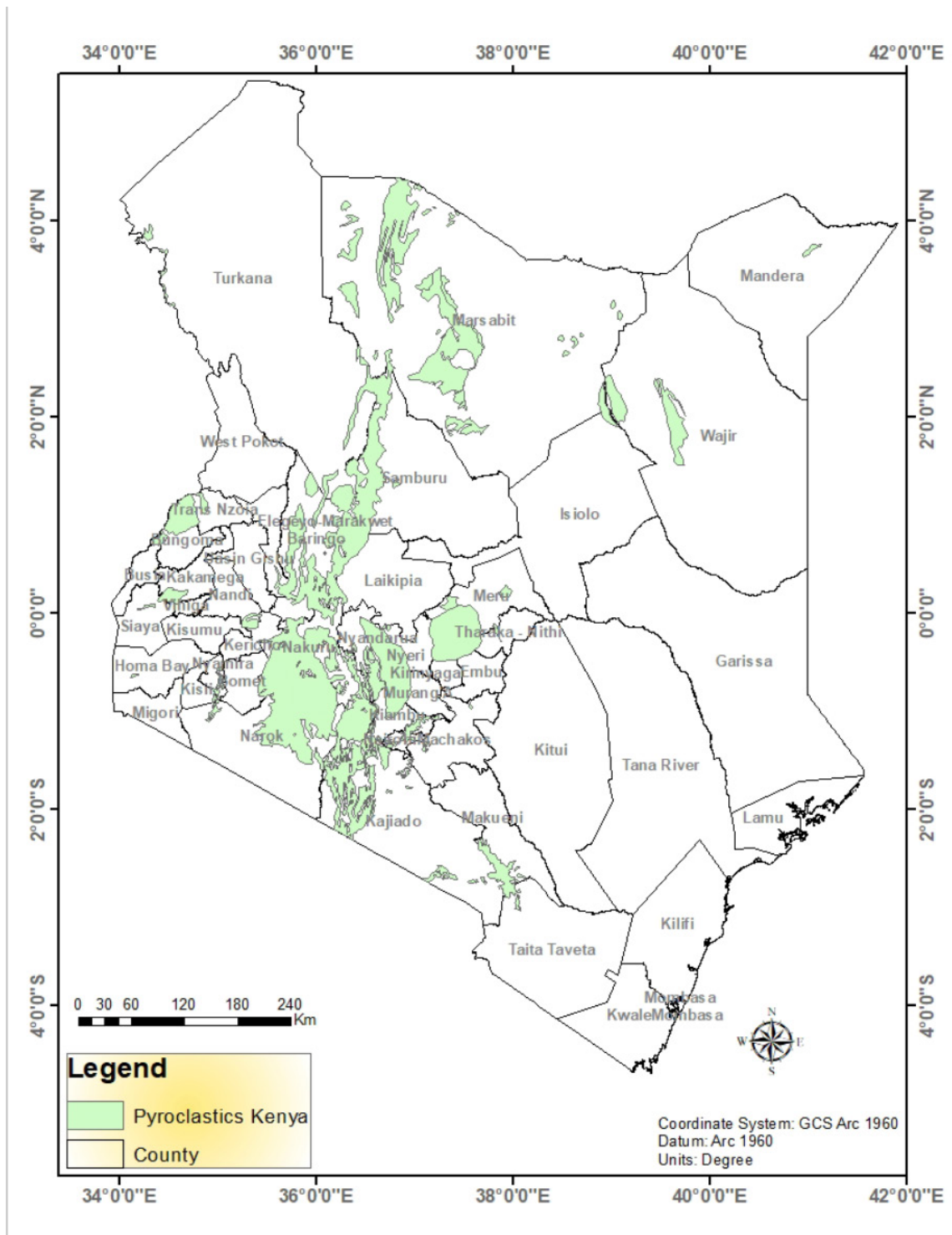
No. 1: Spatial distribution of basalt in Kenya





No. 2 Spatial distribution of granite in the country. It is the most dominant of all intrusive igneous rocks in Kenya.





No. 3 Spatial distribution of pyroclastic in Kenya



6 Legal Framework for Quarrying

There are several pieces of legislation that regulate the quarrying industry, mainly for safety, health and environmental impacts and mitigation. The main ones include:

- a) **The Constitution of Kenya, 2010, Article 61(2)** classifies land as public, community or private land and Article 62(1)(f) declares all minerals and mineral oils as public land. In addition, Article 62(3) provides that all minerals and mineral oils vest in the National Government in trust for the people of Kenya and shall be administered on their behalf by the National Land Commission (NLC). Article 69(1)(a) provides that the State shall ensure sustainable exploitation, utilization, management and conservation of the environment and natural resources and ensure the equitable sharing of accruing benefits. It is on the basis of these provisions of the Constitution that the Mining Act 2016 was enacted.
- b) **The Mining Act, 2016** and its subsidiary legislations which, regulate quarrying with respect to licensing and permitting, mine health, safety and environment, classification of minerals, royalty payments, Community Development Agreements.
- c) **The Environmental Management and Coordination Act (EMCA), 2015** and relevant regulations. They ensure Environmental Impact Assessment (EIA) on proposed projects is carried out and approved. The Act and regulations manage environmental pollution to ensure environmental conservation.
- d) **The Explosives Act, Cap 115** regulates use of blasting explosives to ensure safety of personnel and property, and security of the explosives.
- e) **Occupational Safety and Health Act (OSHA), 2007** ensures safety and health of personnel at workplaces.

6.1 The Mining Act, 2016

The Mining Act is the main statute that regulates mining and quarry operations. The Cabinet Secretary, the Principal Secretary and such other officers may be required, from time to time, to administer the Act. Several subsidiary legislations have been developed in line with the provisions of the Mining Act, to guide the implementation of the provisions of the Mining Act.

The following subsidiary legislations will be referred to for the purposes of this report:

- ▶ The Mining (Licence & Permit) Regulations
- ▶ The Mining (Safety) Regulations
- ▶ The Mining (Royalty) Regulations
- ▶ The Mining (Community Development Agreements) Regulations.

6.1.1 Regulations

6.1.1.1 The Mining (Licence & Permit) Regulations:

The Mining (Licence & Permit) Regulation's subsidiary legislation provides the processes and procedures for acquisition of mineral rights. The Regulations recognize requirements of other pieces of legislations that regulate certain aspects of mining and quarrying. For instance, the application for grant of mineral right requires that the applicant comply with the requirements of the Environmental Management and Coordination (EMCA) Act and any regulations and guidelines. The Environmental and Social Impact Assessment Licence is a requirement for an application for



a mineral right. Section 33(2)(b) provides that the fate of the application should be decided within 120 days, a period which may pose a challenge, especially for quarrying of construction materials to be used in projects with contractual timelines.

6.1.1.2 The Mining (Safety) Regulations Revised 2016):

The regulations give guidelines for safety in mining with respect to extraction, processing; and use, storage and transport of blasting explosives within the mining areas. In this respect section 178(1) of the Mining Act provides that there is no exemption from compliance with the provisions of the Occupational Safety and Health (OSHA) Act, 2017 concerning the safety of mine workers and mine operations.

6.1.1.3 The Mining (Royalty) Regulations:

The subsidiary legislation deals with all aspects of royalty calculation and payments. Section 188 (1) of the Mining Act provides for revision either for reduction or suspension of royalty payment.

6.1.1.4 The Mining (Community Development Agreements) Regulations, 2017

Regulation 4 provides that a community development agreement shall be a legal requirement of the holder of mining licence. The holder is obligated to the community on social and economic contributions. In case that the mining area traverses more than one community, one agreement for multiple affected mine communities will suffice.

6.1.2 Categorization of Quarry Minerals:

Section 2 of the Mining Act provides for the classification of minerals, and a list of all the minerals is given in the First Schedule. The minerals are categorized according to the application, with the first category listing “Construction and Industrial Minerals”.

6.1.2.1 Categorization of Quarry Minerals:

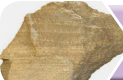
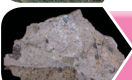
Section 2 of the Mining Act provides for the classification of minerals, and a list of all the minerals is given in the First Schedule. The minerals are categorized according to the application, with the first category listing “Construction and Industrial Minerals”.

6.1.2.2 Classification of Construction Minerals

Section 4 of the Mining Act, 2016 gives the definition of construction minerals -“Construction Minerals include stones, gravel, sands, soils, clay, volcanic ash, volcanic cinder and any other minerals used for construction of buildings, roads, dams, aerodromes and landscaping or similar works, and such other minerals as the Cabinet Secretary may from time to time declare to be construction minerals, by notice published in the Gazette.” In exercise of the powers conferred by Section 4 of the Mining Act, the Cabinet Secretary declared Construction Minerals vide Legal Notice No.62/2017 published in the Kenya Gazette.

The list contains minerals many of which are used for aggregates production. The full list reads as follows:



	Agglomerates		Gravel		Quartzite
	Basalt		Grey whack		Rhyolite
	Breccia		Grits		Sandstone
	Calcrete		Gneisses		Sandstone
	Calc silicate		Kunkar		Siltstone
	Conglomerates		Laterite		Shale
	Dacite		Migmatite		Trachyte
	Diorite		Phyllite		Tuffs
	Dolerite		Phonolite		Soapstone
	Granite		Pyroclastic		Wollstonite
	Granitoids				

6.1.2.3 Categorization of Minerals for Aggregates Production

In order to identify the minerals suitable for aggregates production it is necessary to describe each mineral listed as construction materials in terms of its composition and use.

- ▶ **Agglomerates** – these are tuffs made of coarse accumulation of large blocks of volcanic materials.
- ▶ **Basalt** – it is an extrusive igneous rock widely used in the country for aggregates production. It is majorly used to produce ballast for lightweight engineering structures.
- ▶ **Breccia** – A coarse grained clastic rock composed of angular broken rock fragments held together by a mineral cement of fine grained matrix. It is used as a dimension stone for interior building veneers, tiles, window sill and other decorative applications.
- ▶ **Calcrete** – It's formation resembles that of limestone and is used in road construction.
- ▶ **Calc silicate** – A metamorphic rock consisting mainly of calcium bearing silicates. Like marble, it is used as dimension stone in architectural decorations.
- ▶ **Conglomerates** – A coarse grained sedimentary rock composed of rounded fragments embedded in a matrix of cementing materials such as silica.
- ▶ **Dacite** - Sometimes used to produce crushed stone, it performs well as fill and as loose aggregate in a wide variety of construction projects.
- ▶ **Diorite** – It's durability compares very well with granite and trap rock. It is used as base material in the construction of roads, building and parking areas.
- ▶ **Dolerite** – Used mainly as crushed aggregate in concrete production and as road sub base.
- ▶ **Granite** – It is an intrusive igneous rock, harder than the extrusive igneous rock. It is used as dimension stone for its aesthetic qualities.
- ▶ **Granitoids** – These are granitic rocks.



▶ Gravel – These are smooth rounded pebbles excavated from river beds and are used for decorative purposes on pavements and parking areas.	▶ Rhyolite – Granite's intrusive granite, only that its grain/crystal is relatively smaller. It is commonly mined for ballast for rail and road construction.
▶ Greywacke – A variety of sandstone generally characterized by its hard, used as building stone.	▶ Sandstone – A sedimentary rock used as dimension stone for floor tiles and pavements (Mazeras tiles).
▶ Grit stone/grit - A coarse grained siliceous sandstone used for building or to sharpen blades.	▶ Siltstone – A leader in quartz surfaces for kitchens and baths. It is more expensive than granite.
▶ Gneisses – A high grade metamorphic rock formed from metamorphosis of granite or sedimentary rock used as dimension stone.	▶ Soapstone – A metamorphic rock highly resistant to heat. It is used for fireplace and wood burning stoves. It is widely used to make carvings.
▶ Kunkar – A pebble (gravel) excavated from riverbeds. They are mainly used for decorative purposes and on pavements.	▶ Shale – A mudstone; fine-grained, clastic sedimentary composed of mud that is a mix of clay minerals and tiny fragments of other minerals.
▶ Laterite – It is both soil and rock, rich in iron and aluminium and is commonly considered to have formed in hot and wet areas. Nearly all laterites are rusty and red, because of high iron oxide content. It is cut by a circular saw to make laterite stone tiles for decorating walls.	▶ Slate – A fine-grained foliated metamorphic rock derived from an original shale type sedimentary rock composed of clay or volcanic ash. It is used to make roofing tiles because of its durability and attractive appearance.
▶ Migmatite – A metamorphic rock, which typically contains alternating lighter layers. It contains quartz, feldspar and muscovite. It is used in the construction industry as dimension stone and making natural cement used in the medical industry.	▶ Trachyte – Mainly mined for foundation rocks. It is crushed for ballast for light weight engineering structure.
▶ Phyllite - A foliated metamorphic rock composed mainly of tiny mica grains in parallel alignment. It is not strong enough to serve well as crushed stone. Its slabs are used for landscape, paving or sidewalk stone.	▶ Tuff – A type of rock made of volcanic ash (soft rock) used for building stones used in construction.
▶ Phonolite – The best igneous rock variety for aggregate production. It is best preferred due to its natural resistance to abrasion, especially in road construction. It is also relatively resistant to weathering compared to other extrusive igneous varieties.	▶ Wollastonite – A calcium inosilicate mineral used in automobile brakes, ceramics, paint and plastics. It also serves as an asbestos replacement.
▶ Pyroclastic – The main sources of building stones.	From the description and use for each rock in the list above, it is possible to pick out the list of minerals used for aggregate production:
▶ Quartzite – A metamorphic rock consisting mainly of quartz and is formed by crystallization of sandstone. Used as dimension stone for decorative purposes in houses on walls, flooring and counter tops.	1) Basalt 4) Diorite 7) Phonolite 2) Calcrete 5) Dolerite 8) Pyroclastic 3) Dacite 6) Granite 9) Rhyolite
	However, for the purposes of regulation, all the 31 minerals should be considered together since they are either minerals for aggregate production, gravel, dimension stone or soil. They are therefore development minerals (low value minerals).



7 Quarry Processes

- 1) The factors that are considered when choosing a quarry site include availability of quality rock and distance from construction, which should be close to the site, to minimize costs whilst ensuring it is not close to towns. This is because quarry operations are full of hazards, which include negative environmental impacts such as: Dust from drilling, blasting, crushing or grinding, and moving trucks
- 2) Noise produced from blasting, crushing and other machinery
- 3) Ground vibrations generally from heavy blasting
- 4) Fly rock from poor blasting designs
- 5) Effect on the amount and quality of water
- 6) Lowering the water table.
- 7) Effect on landscape.
- 8) Waste materials.

The other factors that are considered include distance from main roads. The quarry should be close to the main network roads leading to those towns. Other important factors include availability of water for the workforce, dumping space for rock waste and good drainage system to drain both ground and surface water. There are varying methods of quarrying that are determined by the expected end products. They include the following:

- 1) Wedge method where you prepare holes and cracks and use long iron bars inserted in the holes or cracks in order to dislodge or break the rock.
- 2) Channelling method suitable for dimension stone quarrying.
- 3) Quarrying by heating, which is an old and crude method, but may still be used locally to obtain small quantities of stones.
- 4) Drilling and blasting.

7.1 Drilling and Blasting Method

The processes used to recover the rock material will depend on the characteristics of the deposit. The process begins with the removal of overburden to expose the rock deposit ready for drilling. The removed overburden is stored for later use in the restoration process once extraction is finished.

7.1.1 Drilling and blasting

The process begins with a detailed survey of the quarry face to enable the Explosives Engineer to design the blast, which includes where the drill holes are plotted so that the blast can be carried out safely and efficiently. The placement of blasting materials is professionally planned to ensure the required fragmentation is achieved. A single blast can produce up to 60,000 tons of rock.



7.1.2 Crushing

The crushing plants used differ in their production capacity, which determines the size of the operation. Crushing can be done in four stages i.e. primary (first stage), secondary (second stage), tertiary (third stage) and in some quarries a quaternary (fourth stage). After blasting and ensuring that the muck pile is safe, it is transported to the crushing plant where it is fed into the primary crusher. The crushed rock is transported along the process line on conveyor belts or down chutes. Primary crushing is usually done by a jaw crusher consisting of a heavy metal plate. The output from the primary crusher is conveyed to the primary stockpile from which the secondary crusher is fed. The screen house just after the secondary crusher screens out small size crushed stone and dusts onto blinding stockpiles. The larger stones pass through to the final crushing stages where they are fed through a series of cone crushers and screens. The output from the final cone crusher is conveyed to the screen house where large multiple deck screens sort the crushed stone into required aggregate sizes.

7.1.3 Pulverization

Limestone may be further processed to powder for industrial and agricultural use.



8 Cost of Production in Aggregates Quarries

There are 200 aggregates quarries in Kenya with a total production of 60 million metric tons per annum. The average sale price per ton is Ksh 750/= putting gross earnings at Ksh 45 billion per annum.

S/N	Item	Unit	Quantity
1)	Total approximate production per day per quarry	MT	1,000
2)	Number of quarries	Nos	200
3)	Total production per day	MT	200,000
4)	Total production per year	MT	60,000,000
5)	Total gross income per year @ Ksh 750	Ksh	45,000,000,000

% Sale price being spent on taxes, levies, cess & certification	37.15%
% Sale price spent on administrative expenses, establishment & Finance cost	62.85%
Gross income	45,000,000,000
Expenditure	47,555,608,964
Loss	(2,555,608,964)

In considering the cost of production in aggregates quarries the following are considered:

- ▶ Taxes
- ▶ Machinery maintenance
- ▶ Cost of blasting explosives
- ▶ Personal emoluments

Taxes paid by the industry excluding PAYE, Corporate tax, royalties and mining licence/permit fees, include.

Table 2: Taxes, Levies, Cess and Licences

	ITEM	AGENCY	UNIT COST (Ksh) per annum	COST PER TON (KSH)	TOTAL COST (KSH)
1.	VAT @ 16%	National Government (KRA)	103.45 per ton	103.45	6,207,000,000
2.	Quarry cess	County Government	2,000 per trip of 15 MT (4m trips p.a.)	133.33	4,000,000
3.	Quarry extraction fees	National Government	200,000	0.67	800,000,000
4.	Business permit	County Government	86,000	0.29	86,000
5.	Fire permit	National government	8,500	0.03	8,500
6.	Excise duty and road levy	National Government	35 per ton	35	210,000,000
7.	Diesel permit	National Government	18,000	0.06	18,000
8.	Staff work permit	National Government	1,000	3.33	1,000



	ITEM	AGENCY	UNIT COST (Ksh) per annum	COST PER TON (KSH)	TOTAL COST (KSH)
9.	OSHA license	National Government	100,000	0.33	100,000
10.	NEMA license	National Government	100,000	0.33	100,000
11.	Weigh bridge license	National Government	60,000	0.20	60,000
12.	Standard levy	National Government	400,000	1.33	400,000
13.	DIT payment	National Government	24,000	0.08	24,000
14.	NSSF	National Government	48,000	0.16	48,000
Total Taxes, Levies, Cess and Certificates (278.6 per ton)				278.60	16,715,796,551.72

The list of taxes before paying the statutory fees under the Mining Act shows extremely heavy taxation. The counties levy the first three charges on the list, with charges for the quarry extraction and the business permit amounting to double taxation. On the other hand, the charges for extracted minerals are captured as licence fees and royalties charged under the Mining Act.

This therefore means that the county charges do not only constitute double taxation but is also against the provisions of the Mining Act, the Act being the statute responsible for the regulation of mining activities at the national level. It suffices to add that regulation of mining activities is a mandate of the National Government as stipulated in the Constitution of Kenya, 2010, Article 62(1) (f) “all minerals are vested in the National Government in trust of the people of Kenya”.

8.1 Products from Quarrying

A stone quarry typically produces the following products:

- ▶ Large size blocks blasted from the quarry face from approximately 0.5m³ (about 0,36 ton weight) to 2.25m³ (about 5 – 6 ton weight) are called rip rap or rock armour and are used in coastal and river flood defense schemes to shore up sea fronts and river banks.
- ▶ Rubble drawn direct from the muck pile is called face fill and is used as large-scale fill on construction site.
- ▶ Material screened immediately prior to primary crushing is called scalping or grizzly which is again used as fill on construction sites.
- ▶ Screened out fine material from the secondary crusher is called blinding. Some screens have multiple decks and can screen out several grades of blinding. As with crusher run, blinding materials contain a mix of sizes. Blinding, because it is finer than crusher run is used for final shaping up of construction where the sub base is the last unbound layer before coated materials are laid.
- ▶ Screened aggregate (ballast) is used for concrete and mortar in building construction and precast products also used in the construction industry.
- ▶ Screened aggregates mixed with bitumen to form asphalt used for road construction.
- ▶ 32/50 aggregate used for railway bed ballast and as filter media in water treatment plants.
- ▶ Crushed limestone is used as the main raw materials, which together with other minerals such as gypsum, and iron ore are used for the manufacture of cement.
- ▶ Pulverized limestone is used in agriculture to stabilize the soil and for industrial use.

The main market for these products is the construction industry and they make the basis on which the industry thrives.



9 Challenges Faced by the Quarrying Industry

Aggregate quarries are experiencing challenges with respect to operation costs, double taxation and encroachment of quarry safety areas.

9.1 Operation Costs

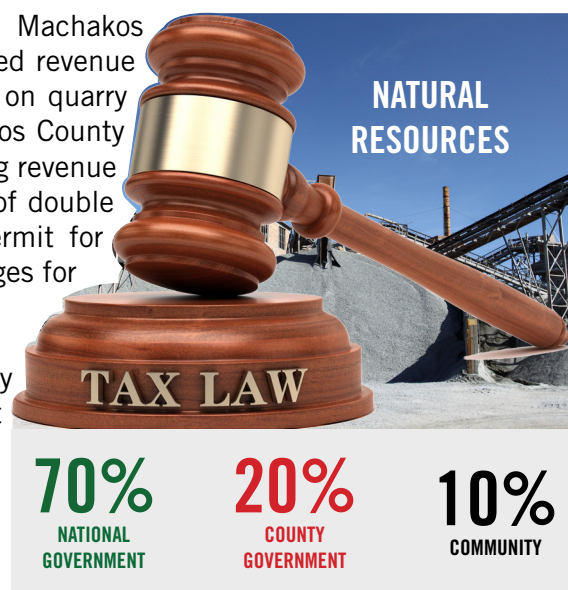
The quarry sub-sector is faced with the high importation costs of regularly replaced machinery parts, such as crushing plant jaw and ball grinders and diamond saws for the dimension stone production.

A reduction in duty paid specifically on the replacement parts shall reduce material costs, which would then support affordable housing.

9.2 Taxation on Quarry Production

There is apparent double taxation by the counties, Machakos County being a case in point. In the quest for increased revenue collection the counties have gone ahead to levy taxes on quarry operations. Machakos County, in enforcing the Machakos County Bill and Machakos Quarrying Activities Act, is collecting revenue from various charges on quarry activities. An example of double taxation on their part, is charging for a business permit for quarry businesses, while at the same time levying charges for a quarrying permit.

Mining, under which quarrying falls, is regulated by Mining Act, 2016 which is administered by the Cabinet Secretary for Petroleum and Mining. Royalty on natural resources is defined as payment received by the owner of real property for exploitation of mineral rights on his property.



In the case of minerals as stipulated in the Constitution of Kenya, 2010, Article 62(1) (f), all minerals are vested in the National Government in trust of the people of Kenya.

The constitution mandates the State to ensure that the accruing benefits from natural resources are equitably shared. The Mining Act, 2016 provides the formula for equitable sharing of the benefits accruing from royalties, the counties being beneficiaries.

The formula for sharing the benefits accruing from minerals provides for 70% for the National Government, 20% for the county holding the minerals and 10 % for the community on whose land the mining activities are carried out.

9.3 Minerals Royalties

Laws have been put in place to manage the revenue flows. For instance in Kenya the royalty payment is regulated by the Mining (Royalty) Regulations, as a mining subsidiary legislation. The regulations provide for different rates for different minerals as has been classified in the first schedule of the Mining Act 2016. The construction and industrial minerals attract royalties of 2% and 1%, respectively. This is to say that aggregates being construction minerals attract royalty payment of 2%.

According to the Working Paper, the levy was set at 1.60 Sterling Pounds per tonne in 2002, increased to 1.95 per tonne in 2008 and 2.00 per tonne in 2009. This levy is mainly for externalities caused by quarry activities. The Annual tax revenues generated from the Aggregate Levy are around 300–350 million sterling Pounds.



9.4 Market Factors that Influence Royalty Payment Negotiations

Royalty payment agreements are fundamentally the result of negotiations between parties involved. The strength of negotiations is influenced by a number of market factors.

- ▶ **Stone quality**” The most desired in Illinois is dolomitic variety (Dolomite) is of high quality.
- ▶ **Land value:**The higher the price of land for uses other than mining, the higher the royalty expected by the leasor.
- ▶ **Quantity of reserves:** The quantity of the rock reserve determines the life expectancy of the mining operation. A larger quantity per acre extends the life span of the mining operations at the site and makes investment more economically viable.
- ▶ **Market condition:** Proximity to stone markets for construction and the demand size have a major influence on royalty payments.
- ▶ **Location:** Quarry locations in relation to the market determine how much transport cost will have to be paid by the customer. Quarry locations closer to the market thus have competitive advantage that could affect the producer's willingness to pay a higher royalty.

9.5 Royalty Agreements in the United States Crushed Stone Industry

There are two forms of agreements in the US where 2/3 of (the States) of the agreements are in US dollars per ton while the remaining 1/3 of the agreements are for payment in percentage of sales price. In general, the outright purchase agreements seem to pay higher royalty rate than the per ton agreement.

Royalty payment in the US vary from State to State, and ranges from US \$ 0.08 -1.00 per tonne; and 3 % - 11.25 of market price (**ISGS Guidebook 17, 1985**).

9.6 Zoning of Quarry areas

Zoning is defined as a well-established planning tool used to divide areas of land into areas to manage particular effects, activities or uses

within the area through application of rules, including performance standards. This definition was adopted in the guidance note initiated by the Aggregate and Quarry Association of New Zealand (AQANZ) in partnership with the Ministry for Environment and Local Government, New Zealand. The main focus of the guidance note is planning and management of medium and large scale quarrying of aggregates.

Zoning can be used to identify potential aggregate extraction sites and to protect established sites from incompatible or sensitive activities, which may lead to conflicts. Zoning can also be effective to discourage quarries in areas valued for other areas as ecology, natural character or amenity.

In some instances, it may be necessary to develop a hierarchy of zones or policy areas to provide areas for the appropriate level of protection of access to aggregate resources such as:

- ▶ A quarry zone where the quarries are an established activity, with plan objectives, policies and rules allowing and encouraging quarry development and/or discouraging other activities from establishing in that zone.
- ▶ A potential aggregate extraction where a combination of factors indicates that aggregate extraction could be appropriate in the future.
- ▶ A buffer zone around a quarry site where activities are more tightly restrained to serve as a reminder of the potential for adverse effects from the quarry within the quarry zone
- ▶ Although the National Land Use Policy (Draft 2016) does not expressly provide for zoning of quarry areas, it does appreciate the problems of an urban environment with respect to peri-urban mining, quarrying and other extractive activities. It states that, for the Government to mitigate the problem, it should establish measures to regulate the industry. NEMA's Integrated National Land Use Guidelines, 2011 pointed out key land use challenges and threats, while giving guidelines for mining and quarrying. The land use challenges in Kenya include: Incompatibility and land use conflict
- ▶ Conflicting sectorial land use statutes



The guidelines for mining and quarrying recognize the importance of quarry industry in supporting the construction industry, and create huge employment opportunities. However, there are concerns about the negative environmental impacts of quarry activities, which need to be addressed. The guidelines are broadly categorized into three namely:

- a) Occupational, public safety and health guidelines
- b) Environmental guidelines
- c) Socio-economic guidelines

Under (a) above safety distances have been prescribed but only for quarry operations without blasting, thereby not addressing the negative effects of blasting on the environment, and that blasting of quarries should be in harmony with other land uses, as well as provide for defined buffer zones between quarries and other land users.

9.7 Quarry Buffer Zones

A buffer zone is a neutral zonal area that lies between two or more bodies of land, usually pertaining to countries. Depending on the type of buffer zone, it may serve to separate or cojoin them. A buffer zone so created for quarrying is either to protect the quarry from eventual encroachment to the quarry safety area from other land uses or to protect life and property in close proximity to quarries. Quarries are commonly located either in remote areas that are sparsely populated or in close proximity of settlements. In order for the quarry to start operations, there is need to consider supporting infrastructure for instance power, access roads and adequate water. The availability of these provisions acts

as the motivation for other developments to spring up in the vicinity. This eventually creates conflict between quarrying and the neighbouring developments brought by the negative impacts associated with quarrying.

Quarrying produces negative environmental impacts on life and property in proximity. These impacts include:

- ▶ Dust from drilling, blasting, crushing or grinding, and moving trucks
- ▶ Noise produced from blasting, crushing and other machinery
- ▶ Ground vibrations generally from heavy blasting
- ▶ Fly rock from poor blasting designs
- ▶ Ground water quality

The degree and nature of these effects caused by quarrying vary according to the type of quarry, scale of operation, methods used to excavate the aggregates, the geology of the area, the receiving environment and the surrounding land uses. Although these negative environmental impacts can be mitigated, they cannot always be avoided.

When buffer zones are required for quarry operations, buffers establish an area around the existing quarry or activity zones that prevent those that are sensitive to quarrying, located in the area.

In considering the use of buffer zones, the regulating authority should be satisfied that the effects from quarrying are internalized, as far as is reasonable and consider the appropriate distance to mitigate the effects in question against the significance of the quarrying activity.





10 Conclusion and Recommendations

10.1 Conclusion

Aggregate quarrying is an integral part in economic development contributing immensely in infrastructural development and construction in general. Quarrying enterprises, like any other business concerns, experience operational challenges, which could be resolved. The challenges and other concerns expressed by aggregate quarries in the county include heavy taxation, taxation by County Governments despite the fact that regulation of mining/quarrying is a function of the National Government. The problem of encroachment to the quarry safety areas by incompatible activities is a challenge as well as a concern to the industry. This if not checked would result in relocation of the quarries as has happened in the past, creating job losses not to mention that the quarries suffer damages as a result of closure and relocation.

10.2 Recommendations

It is recommended that the Quarry Sub-sector, together with the Ministry of Petroleum and Mining and possibly the Council of County Governors, resolve the issues of concern. The Quarry Industry needs to petition the Cabinet Secretary for Mining to establish a task force comprising line Ministries and other relevant stakeholders to deliberate on the concerns with a view to creating a conducive business environment for the quarries.



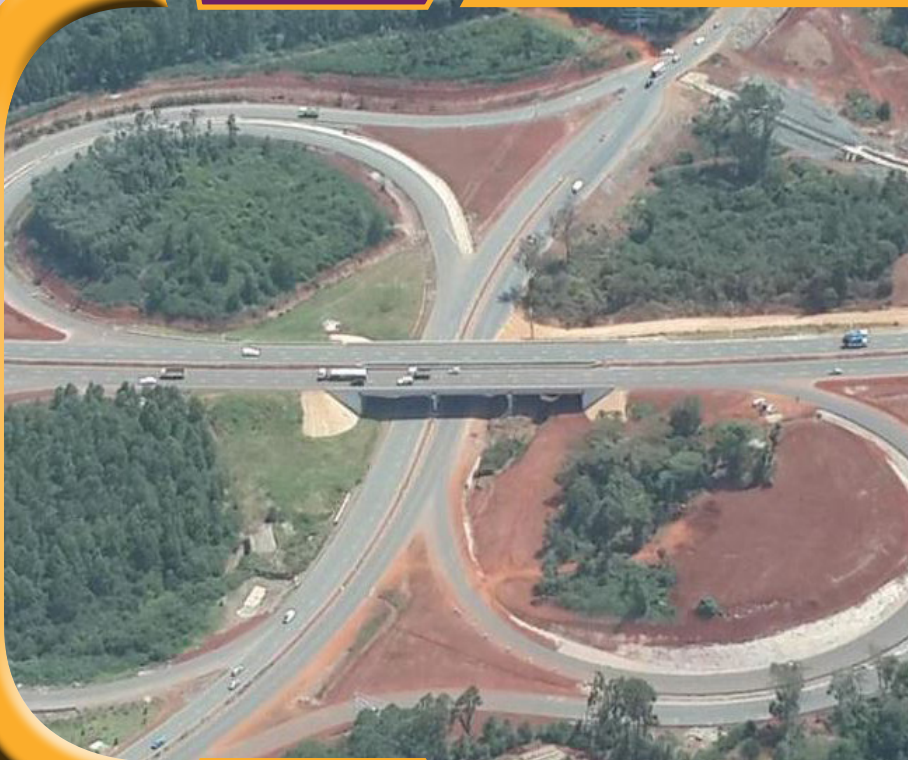


S/N	Issue	Comment	Recommendation	Key stake holders group
1)	Mining Act(2016) review	Quarry products are low value products, Howeverm they are considered high value minerals under theMining.. The end products from the Quarry sector are aggregate materials and not minerals.	Review the Mining Act(2016) to classify quarry products as low value products.This will help harmonize and rationalize payment of fees prescribed in the Act.	Ministry of Petroleum and Mining
2)	Royalty fees	The act requires a holder of a mineral right to pay royalties to the State in respect of the various mineral classes won by virtue of the mineral right. (2) The Cabinet Secretary shall prescribe the rates payable under subsection (1). (3) The Cabinet Secretary may require the holder of a mineral right to make returns relating to any royalties paid in such manner and within a period as may be prescribed. (4). Any mineral samples including core samples, removed for the purposes of testing, shall not be subject to royalty unless they exceed maximum value stipulated in regulations. The construction and industrial minerals attract royalties of 2% and 1%, respectively. This is to say that aggregates being construction minerals attract royalty payment of 2%.	Further, royalty payment for construction materials could be calculated using the market factors that influence royalty payment negotiations (see below Royalty payment agreements are fundamentally the result of negotiations between parties involved. The strength of negotiation positions is influenced by a number of market factors. Stone quality – the most desired in Illinois is dolomitic variety (Dolomite) is of high quality ▲ Land value; the higher the price of land for uses other than mining, the higher the royalty expected by the lessor ▲ Quantity of reserves; the quantity of the rock reserve determines life expectancy of the mining operation. Larger quantity per acre extends the life span of the mining operations at the site and makes investment economically more viable. ▲ Market conditions; proximity to stone markets for construction and the demand size have a major influence on royalty payments ▲ Location; quarry locations in relation to the market determine how much transport cost will have to be paid by the customer. Quarry locations closer to the market thus have competitive advantage that could affect the producer's willingness to pay a higher royalty.	Ministry of Petroleum and Mining

S/N	Issue	Comment	Recommendation	Key stake holders group
3)	Cost of electricity	Electricity costs account for a large proportion of the conversion costs for the quarry companies leading to lower gross profit margins on sales. High-power costs and fluctuations have a high impact on business competitiveness. Further, the electricity tariff could be reviewed to ensure the competitiveness of locally manufactured products.	Review electricity rates to at least USD 0.09 per kWh to quarry companies. This will enhance industry competitiveness. To address power fluctuations and quality, there is need to build dedicated sub-station close to the quarry zones.	Kenya Power
4)	Import duty and VAT on Machinery and Machinery parts	In 2020, Government introduced VAT on imported machinery (16%) making the cost of production machines very expensive.	Zero rate machines.	National Treasury
5)	Double taxation	There is need to address national and county governments' duplicating regulatory roles, which result in the high costs of doing business. For example County Governments are now developing Acts that have similar mandates to the Mining Act. For example, the Machakos County Management of Quarrying Activities Act, 2016 which seeks to "regulate quarrying activities; to ensure sustainable exploitation and utilization of quarries; to provide for raising of revenue from quarrying activities; to provide for equitable sharing of the accruing benefits;"	Development of legal framework to guide taxation structure in the country of each level of government. This will ease doing business and remittance of taxes of quarry operations without any conflicts.	Ministry of Mining and Petroleum and Council of Governors and National Treasury
6)	Zoning quarrying areas	Absence of clear demarcation or zoning of industrial and residential areas resulting in numerous conflicts between Industries, Residents/communities, and Provincial Administration. The need to create zones for quarries is essential for safety of communities due to the nature of their operations.	Set a Task Force to come up with clear guidelines on zoning quarrying areas, creation of buffer zones and creation of protected areas	Ministry of Petroleum and Mining and Council of Governors and Ministry of Lands
7)	Skills development	The government should consider a technological skills cooperation with already technically advanced nations so as to borrow the needed concepts to align our industries within the global dynamics.	Establishment of technical innovation Institute's hubs	TVET Ministry of Petroleum and Mining







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